

Before establishing a BTP session, the GATT client SHOULD perform a GATT Exchange MTU procedure.

After that the BTP Client SHALL execute the GATT discovery procedure. The GATT discovery procedure starts with primary service discovery using either the GATT Discover All Primary Services sub-procedure or the GATT Discover Primary Services by Service UUID sub-procedure.

The BTP Client SHALL perform either the GATT Discover All Characteristics of a Service sub-procedure or the GATT Discover Characteristics by UUID sub-procedure in order to discover the C1 and C2 characteristics.

The BTP Client SHALL perform the GATT Discover All Characteristic Descriptors sub-procedure in order to discover the Client Characteristic Configuration descriptor (CCCD) of C2 characteristic.

To initiate a BTP session, a BTP Client SHALL send a BTP handshake request packet to the BTP Server via a ATT_WRITE_REQ PDU on characteristic C1 of the BTP Service. The handshake request packet SHALL include, a list of BTP protocol versions supported by the client, the client's GATT ATT_MTU, and the client's maximum receive window size. The list of supported protocol versions SHALL be sorted in descending numerical order. If the client cannot determine the BLE connection's ATT_MTU, it SHALL specify a value of '23' (the minimum ATT_MTU supported by GATT) for this field in the handshake request. For a detailed specification of the handshake request binary data format, see [Section 4.19.3.1, "BTP Handshake Request"](#).

After the BTP Client acknowledges delivery of the handshake request packet, upon receipt of a GATT Write Response, the BTP Client SHALL enable indications over C2 characteristics by writing value 0x01 to C2's Client Characteristic Configuration descriptor as described in [Bluetooth® Core Specification 4.2 \(amended\)](#) Vol 3, Part C, Section 10.3.1.1 "Handling of GATT Indications and Notifications".

Once the GATT server has received a client's BTP handshake request and confirmed the client's subscription to C2, it SHALL send a BTP handshake response to the client via a ATT_HANDLE_VALUE_IND PDU on C2. This response SHALL contain the window size, maximum BTP packet size, and BTP protocol version selected by the server. For a detailed specification of the handshake response binary data format, see [Section 4.19.3.2, "BTP Handshake Response"](#).

The server SHALL select a window size equal to the minimum of its and the client's maximum window sizes. Likewise, the server SHALL select a maximum BTP Segment Size for the BLE connection by taking the minimum of 244 bytes (the maximum characteristic value length of C1 and C2), server's ATT_MTU-3 and ATT_MTU-3 as declared by the client.

The server SHALL select a BTP protocol version that is the newest which it and the client both support, where newer protocol version numbers are strictly larger than those of older versions. The version number returned in the handshake response SHALL determine the version of the BTP protocol used by client and server for the duration of the session.

If the server determines that it and the client do not share a supported BTP protocol version, the server SHALL close its BLE connection to the client. When a client sends a handshake request, it SHALL start a timer with a globally-defined duration of BTP_CONN_RSP_TIMEOUT seconds. If this timer expires before the client receives a handshake response from the server, the client SHALL close the BTP session and report an error to the application. Likewise, a server SHALL start a timer